



Commissioning Manual

S50-E6 Series

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Preface	1
Chapter 1 Preparation work before commissioning	2
Chapter 2 Self-learning of the traction machine	2
1. Correct Motor Parameters	3
2. Steps of motor self-learning	3
Chapter 3 Commissioning at inspection speed	5
1. Inspection and Confirmation of Mechanical Assembly	5
2. Inspection and Confirmation of Electrical Assembly	6
3. Power on Inspection	7
4. Adjustment of Door Opening and Closing	9
5. Emergency Operation in Machine Room	9
6. Car Top Inspection Operation	9
7. Pit Inspection Operation (No need to confirm it if there is no pit)	10
Chapter 4 Commissioning at normal speed	11
1. Re-inspection and Re-confirmation of Mechanical Assembly on Site	11
2. Re-inspection and Re-confirmation of Electrical Assembly	11
3. Elevator Shaft Self-learning	12
4. Commissioning method of serial calling board	13
Chapter 5 Elevator Commissioning	15
1. Automatic Leveling Function Operation	15
2. Single Floor Operation	15
3. Multi-floor Operation	15
4. Confirmation of Door Opening and Closing Operation	16
5. Confirmation of Instruction Registration and Self-protection Function	17
6. Commissioning of Elevator Duplex Control Function	17
7. Confirmation of Elevator Balance Coefficient	18
8. Confirmation of Elevator Automatic Evacuation Device Function (no need to confirm if it is not provided)	19
9. Elevator opposite door commissioning (no need to set if there is no this function)	19
10. Instructions of Asynchronous Motor Auxiliary Brake (no need to set if there is no this function)	21
11. Door Lock Bypass Function	21
12. Elevator Car Unintended Movement Detection Function Test (no need to set if no this function)	22
13. Machine Braking Force Monitoring Function Test (no need to set if no this function)	23
Chapter 6 Adjustment of elevator leveling and comfort	25
1. Elevator Leveling Adjustment	25
2. Elevator Starting Comfort Adjustment	25
3. Elevator Operation Comfort Adjustment	25
Appendix 1 Instructions for commissioning tools	26
1. Hand-held Operator Diagram	26
2. Button Function Instruction of Hand-held Operation Panel Keyboard	26
3. Function and Instructions of small keyboard	27
Appendix 2 Controller Function Parameters Description	31
Appendix 3 Fault Information	39
Appendix 4 Adjustment of Door Machine Parameters	46

Preface



This commissioning manual is only applicable to S50-E6 series electrical schematic diagram with the speed range of 0.25m/s ~ 4. 0m/s.



Danger

1. Don't put the combustibles near the elevator control cabinet, otherwise there is a danger of fire;
2. Don't install in an environment with explosive gas, otherwise there is a danger of fire;
3. Don't dismantle the control parts at will, otherwise the system will be damaged;
4. It is strictly prohibited to install near water pipes and in other occasions where there is possibility of water splash. Otherwise, the controller will be damaged;
5. Don't drop screws, gaskets and metals into the elevator control cabinet, otherwise there is a risk of damage;
6. Please don't touch the components of the controller by hand, otherwise the controller will be damaged by static electricity;
7. Please don't touch the controller with wet hands, otherwise there is danger of damage to the controller;
8. Carry out wiring and other operations after confirming that the input power is completely disconnected, otherwise there will be the risk of electric shock;



Attention!

1. The power supply terminal and the wire must be firmly connected, otherwise there will be a danger of damage to the controller;
2. Don't touch the controller terminal or control printed circuit board under the condition of power on, otherwise there will be a danger of damage to the controller and electric shock;
3. The control cabinet housing, frequency converter grounding end, motor housing and car body must be reliably grounded;
4. The shielding line of rotary encoder must be reliably grounded, otherwise it will affect the elevator comfort and cause motor noise, and the elevator may not run normally;

Chapter 1 Preparation work before commissioning

1. At least one person in the commissioning personnels can read all the files of the elevator;
2. The commissioning personnels must be familiar with the principle, structure and operation method of variable voltage variable frequency speed regulation elevator;
3. The commissioning personnels must be familiar with the control principles of the integrated control system;
4. The commissioning personnels must be good at fault judgment, analysis, and elimination;
5. Commissioning personnels must be serious, responsible, and strict, and do a good job of safety in the commissioning process;
6. The instruments needed for commissioning (in general condition)
 - a) Multimeter
 - b) Clamp meter
 - c) Ruler
 - d) Commonly used electrical tools

Chapter 2 Self-learning of the traction machine

The machine room is in the emergency electric state, short-circuit the wire terminals in the control cabinet: 304 and 393, 320n (n is the number of floor) and 320, 320 and 319, 401 and 820. Plug in the commissioning plug JD2, and confirm that the JD6 and the UCMP-T plugs are connected. Confirm that the encoder cable is properly connected, and then connect well the general power supply of the distribution box in machine room and the power line of the control cabinet (L1, L2, L3, N, PE), connect the traction machine and the power lines (U, V, W, PE) of the control cabinet, and connect the brake coil and lines of the control cabinet (501, 502, PE), and connect 401, 816, 803, 803A lines.

1. Correct Motor Parameters

Function code	Name	Setting	Unit
F1-00	Encoder type	0	
F1-01	Rated power	According to motor's rated power	KW
F1-02	Rated voltage	According to motor's rated voltage	V
F1-03	Rated current	According to motor's rated current	A
F1-04	Rated frequency	According to motor's rated frequency	Hz
F1-05	Rated rotate speed	According to motor's rated rotate speed	Rpm
F1-12	Number of pulses per turn of encoder	According to encoder's actual pulse number	
F1-25	Motor type	0: Asynchronous motor 1: Synchronous motor	

2. Steps of motor self-learning

2.1 Use hand-held Operator to do motor self-learning

2.1.1 Press the red button of the operator to reset the current fault. Set F1-11 to 1, and the mainboard LED displays "TUNE". Press maintenance upward or downward button, and the motor first makes a distinct electromagnetic sound, and then operate in accordance with the given direction until the original point signal of the encoder is detected. The motor tuning is completed when the mainboard LED no longer displays "TUNE". After that, the system will be prohibited running for 8 seconds for storage parameters. Do self-learning more than 3 times and compare the encoder initial angle of the F1-06. The error should be within the range of 7 degrees. After learning every time, the values in F1-08 should be the same.

2.2 Small keyboard for motor self-learning:

2.2.1 Set the parameters within the F-2 to "1" (reset current fault) through the small keyboard on the mainboard, and set F-A to "1" through the small keyboard on the mainboard, and the mainboard LED displays "TUNE". Press maintenance upward or downward button, and the motor first makes a distinct

electromagnetic sound, and then operate in accordance with the given direction until the original point signal of the encoder is detected. The motor tuning is completed when the mainboard LED no longer displays "TUNE" and a numerical value (the value is the angle value of the encoder self-learning) appears. After that, the system will be prohibited running for 8 seconds for storage parameters.

- 2.3 After the tuning is completed, carry out the maintenance trial operation, and observe whether the current is normal, whether elevator operation is stable, and whether the running direction is consistent with the given direction. Confirm whether the change of F4-03 pulse is normal (up running increasing and down running decreasing) if there is a operator. If the elevator running in the opposite direction or pulse change is abnormal, please change F2-10 parameters to change the elevator running direction or pulse direction by the hand-held operator. If there is no handheld operator, set the F-C parameter through the small keyboard, so that the system does not report E38 fault.



Prompt:

- 1. The above self-learning operation mode of the motor is only suitable for synchronous motors!**
- 2. Please carry out motor encoder angle self-learning for many times. The error value of encoder angle should be within 7 degrees after self-learning!**

Chapter 3 Commissioning at inspection speed

1. Inspection and Confirmation of Mechanical Assembly

Warning

Before checking, confirm that the power switch of the elevator distribution box and the switch of the lighting power supply should be in the state of disconnection and that all the switches in the elevator control cabinet should be disconnected.

- 1.1. Confirm that there is no obstacle to the elevator operation in the shaft;
- 1.2. Confirm that the components of the elevator pit should be well installed. If the oil buffer is used, the buffer oil must be added as required, and the pit should be kept dry;
- 1.3. Confirm that the landing door and car door are well installed, door vane is installed correctly, the open/close of the door is flexible, and the space between landing door and the door hole is well sealed, and there are necessary protective measures and obvious warning signs outside the landing door to prevent unauthorized personnel into the wells;
- 1.4. Confirm the installation and adjustment of the guide rail are completed and pass the test;
- 1.5. Confirm the correct installation of car door machine;
- 1.6. Confirm the correct installation and fastening of the car wire rope;
- 1.7. Confirm that the speed governor is in line with the national standard and the action is effective;
- 1.8. Confirm the correct installation of speed governor, wire rope and tension pulley;
- 1.9. Confirm that the shaft limit switch has been adjusted to ensure safety and effectiveness;
- 1.10. Confirm that the traveling cable has been installed and reliably fixed and is without knot;
- 1.11. Confirm that the cable ducts are well laid, and there are copper sheets or yellow green wires to connect the ducts;
- 1.12. Confirm that the safety windows (if it is equipped) and the safety gears are correctly installed and act effectively;
- 1.13. The traction machine is fixed in accordance with the requirements of the installation instructions; If the traction machine is a gear traction machine,

check whether the oil level of the reducer is normal;

- 1.14. Confirm that the encoder on the traction machine is securely fixed;
- 1.15. All installations shall be carried out in accordance with the national installation code GB10060;
- 1.16. After the installation of all components is complete, the machine room and shaft must be kept clean.

2. Inspection and Confirmation of Electrical Assembly

Attention

In the following inspection process, any problems must be carefully examined, and troubleshooting should be carried out to confirm that it will not affect the equipment and personal safety.

- 2.1. Confirm that the power cables L1, L2, L3, N and PE between distribution box of the elevator and the control cabinet are connected;
- 2.2. Confirm that the brake coil of traction machine has been connected to 501, 502, 401, 803, 401 and 824 in the control cabinet;
- 2.3. Confirm that the control cabinet power cables U, V, W, PE have been connected to power cables U, V, W, PE of the traction machine;
- 2.4. Confirm the connection between encoder and inverter;
- 2.5. Confirm that all the switches of safety circuit components have been reset; confirm that the control cabinet safety circuit 304 and 320+n have been connected, and if not, troubleshooting must be taken to solve problems; short circuit safety circuit should be prohibited;
- 2.6. Confirm that the door lock circuit 320+n (n is the floor number), 320 and 319 have been connected, and if not, troubleshooting must be taken to solve problems; short circuit safety circuit in the cabinet should be prohibited;
- 2.7. Confirm the correct connection in the car top inspection box and correct on-off logic of the car top inspection circuit;
- 2.8. Make sure that the lighting circuit is connected correctly, no short circuit or broken circuit;
- 2.9. Confirm that the connection of the door machine is correct and that the door

machine power supply connection and signals are correct;

- 2.10. Confirm the correct wiring of the operation panel;
- 2.11. The insulation resistance between the grounding wires and the terminals (including power line and control line) should meet the requirements of the national standard. If there is any discrepancy, inspection should be carried out immediately;
- 2.12. The connection point of the grounding wire should be concentrated in the grounding signs of the control cabinet and must not be dispersed, and the grounding should be firm without any loose phenomenon;

3. Power on Inspection

- 3.1. Confirm that the inspection switch in the control cabinet is in the emergency electric state, and the inspection switch on the top inspection box is in normal condition;
- 3.2. Check the power incoming voltage of three-phase five-wire power supply. The three-phase voltage should be $380\text{VAC} \pm 7\%$, and the voltage between each phase and zero line should be $220\text{VAC} \pm 7\%$;
- 3.3. When the power supply is normal, turn on electricity to do the commissioning;
- 3.4. Turn on the power switch, the "Normal" normal light of the control cabinet power protector (JXW) is on; if the "Alarm" indicator light is on, it indicates that the input three-phase power supply has the faults of wrong phase or lacking phase. Thus, please turn off the power and exchange any two phases of the three-phase lines or carry out commissioning after excluding other fault;
- 3.5. Check all safety contact switches: under normal conditions, the main board X25 input signal should be effective, and then disconnect safety protection switch one by one. The safety protection switches that can disconnect the X25 input signal of the main board:

Code	Name	Code	Name
KTJ	Car safety window switch	KSB	Speed governor tension pulley switch
KXJ	Car safety gear linkage switch	XJX	Shaft lower limit switch
ZT	Stop switch of car top inspection box	SJX	Shaft upper limit switch
KDT1	Pit stop switch 1	KSV	Over speed switch of governor
KDT2	Pit stop switch 2	KFS	Machine room stop switch
JHK	Car buffer switch	KNT	Car stop switch
DHK	Counterweight switch	KHW	Handwheel switch
KNR	Rope gripper switch	JX	Limit switch (for car)
KDC	Positioning switch for car inspection	KDM	Traction machine side stop switch

3.6. Check whether the voltage data of the control cabinet terminal is normal:

Control cabinet terminal	301-303	703-704	753-754	401-402
Measured voltage (V)	AC110 $\pm$ 7%	AC220 $\pm$ 7%		DC24V $\pm$ 7%

3.7. Check the control board input light:

Component description	Component code	The state of I/O when it is acting
Leveling switch	YMQ-A, B	When leveling X2——on, synchronous machine without YMQ-B
Upper leveling switch	YMQ-S	When leveling X1——on
Lower leveling switch	YMQ-D	When leveling X3——on
Door lock switch	KMM	Pull-in X5——on
Operating contactor	KMY	Pull-in X6——off
Braking contactor	KMB	Pull-in X7——off
Star-delta contactor	KMYF	Pull-in X8——on
Machine room emergency operation、Car top inspection switch	KAG,KAD	During maintenance X9——off, it is on when it is in automatic state.
Inspection upward	ADS,AGS, APS	During action X10——on
Inspection downward	ADX,AGX, APX	During action X11——on
Pit inspection switch	KAP	During action X12——off
Upward single floor forced deceleration	SXY	During action X14——off
Downward single floor forced deceleration	SXD	During action X15——off
Upward multi floor	SSY	During action X16——off

Component description	Component code	The state of I/O when it is acting
forced deceleration 1		
Downward multi floor forced deceleration 1	SSD	During action X17——off
Upward multi floor forced deceleration 2	SSY1	During action X22——off
Downward multi floor forced deceleration 2	SSD1	During action X4——off
Safety circuit		When safety circuit is on, X25—on (high voltage input)
Door lock circuit		When landing door is closed, X26—on (high voltage input)
Door lock circuit		When car door is closed, X27—on (high voltage input)

4. Adjustment of Door Opening and Closing

Adjust the relevant parameters according to the commissioning instructions of the door assembly to ensure that the door lock circuit acts effectively. Adjust the parameters as appendix four;

5. Emergency Operation in Machine Room

5.1. Must ensure that there is no person in the shaft and on top of the car before emergency operation in the machine room;

5.2. The car top and pit inspection switches (KAD, KAP) are in normal condition, and the machine room inspection switch (KAG) is in the emergency state. Connect JD6 short connector when there is no pit inspection box;

5.3. Press the inspection upward (AGS) or inspection downward (AGD) running button and confirm (KNG) button on the inspection box in the machine room, and the elevator starts to run;

6. Car Top Inspection Operation

6.1. The car top inspection operation can be done when the pit inspection is in the normal mode. Before car top inspection operation, the car top inspection circuit should be confirmed to be correct, and each button's function is normal, especially the car top emergency stop switch function.

6.2. During the operation, if it is found that the inspection upward and downward

buttons of the car top are inconsistent with the actual running direction of the elevator, the inspection circuit of the car top should be checked, but the wiring of the main circuit can not be replaced or changed in the control cabinet.

7. Pit Inspection Operation (No need to confirm it if there is no pit)

7.1. The pit top operation can be done when the car top inspection is in the normal mode. Before pit inspection operation, the pit inspection circuit should be confirmed to be correct, and each button's function is normal, especially the pit emergency stop switch function.

7.2. During the operation, if it is found that the inspection upward and downward buttons of the pit are inconsistent with the actual running direction of the elevator, the inspection circuit of the pit should be checked, but the wiring of the main circuit can not be replaced or changed in the control cabinet.



Prompts:

- 1. Please follow the principle of priority of car top inspection or pit inspection operation and machine room emergency operation!**
- 2. Please follow the interlocked operation principle of the car top inspection and pit inspection operation!**
- 3. Starting the inspection or emergency operation after the buzzer in the car or the alarm at the car bottom rings continuously for three seconds, and the buzzer or the alarm will ring intermittently during inspection operation!**

Chapter 4 Commissioning at normal speed



Note: The following commissioning must be carried out after the elevator is installed.

1. Re-inspection and Re-confirmation of Mechanical Assembly on Site

Warning

Before the commissioning at normal speed, the commissioning at inspection speed must be carried out; The commissioning at normal speed must be carried out after the commissioning at inspection speed is qualified.

- 1.1. Outside the hall shall be with necessary protective measures and obvious warning signs to prevent unauthorized personnel from entering the elevator;
- 1.2. The upper and lower limit switches of the shaft have received necessary adjustments to ensure the safety and effectiveness;
- 1.3. Confirm the correct installation of upper and lower forced deceleration switches;
- 1.4. When the speed of the elevator is not more than 1.75m/s, it is required to install corresponding limit switches and single floor forced deceleration switches in the shaft, 4 pieces of switches in total, and for the elevators with other speed, please refer to the electrical schematic diagram and elevator installation documents;
- 1.5. Confirm that the installation position of each floor leveling board is right;
- 1.6. Confirm that the car travelling cable is installed and is fixed and reliable, no knotting phenomenon;
- 1.7. Safety window (if it is provided), the safety gears are installed correctly with effective action;

2. Re-inspection and Re-confirmation of Electrical Assembly

Notice

In the following inspection process, any problems must be carefully examined, and troubleshooting should be carried out to confirm that it will not affect the equipment and personal safety.

- 2.1. Re-confirm that the three-phase five wire power supply voltage which is supplied to the control cabinet is normal;
- 2.2. Confirm that the safety circuit 304 to 320+n is connected, an short circuit safety circuit is strictly prohibited;
- 2.3. Confirm that the door lock circuit 319, 320, 320+n (n is the floor number) have been connected, and short safety circuit should be prohibited except door lock bypass function ;
- 2.3. Confirm that the wiring of operation panel and the door machine are correct;
- 2.4 Confirm that the door opening and closing operation is normal.

3. Elevator Shaft Self-learning

- 3.1 The elevator must be in the state of inspection mode, and must set F6-00 (the maximum floor number of the elevavor) in the controller correctly;
- 3.2 Make the elevator run at inspection speed to the door zone position of the lowest floor, and the levelling switch acts. Must ensure that the forced deceleration switch is acting at the same time, and the current floor number shall be 1 (monitoring F4-01);
- 3.3 Enter F-7 mode through the keys UP and SET on the main control board, and change the data to 1, and then change F1-11 to 3 by pressing the key SET or by hand-held operator. The system automatically performs shaftl floor height self-learning, and the elevator will automatically run at inspection speed to the leveling position at the top floor.
- 3.4 In the course of shaft self-learning, when reaching each leveling floor, the levelling switch signal light X2 will be on once, and the displayed floor on the digital tube will automatically add 1;
- 3.5 When the elevator automatically runs to the leveling position of the top floor, the floor height self-learning of the shaft is completed, and the system will automatically withdraw from the shaft self-learning mode. When fault occurs during self learning, the current fault code will be displayed on the mainboard digital tube, and please refer to the appendix in detail.



1. In the shaft self-learning process, the elevator must be under inspection or machine room emergency

operation condition, and the door lock and safe circuit shall not be disconnected. Otherwise it will lead to the failure of self-learning!

2. When the elevator floor number is 2, and the elevator is under shaft floor height self-learning, firstly the elevator must be break away from the door zone position of bottom floor!

4. Commissioning method of serial calling board

4.1 Non-ultra-thin calling board setting method:

Press the button S1 on serial board of the calling board, and it will show the set floor address. Then press the button again and the floor address will plus 1; each time press the button, the floor will add 1, and if press S1 for a long time, the floor address will add 1 continually, and the maximum can be set to 40. If the button is not pressed in the continuous 3S, the current data is stored as the floor address of the display board. If the floor data flashes twince indicating that the storage is successful, and the button is invalid during the flashing period.

4.2 Ultra-thin calling board setting method:

Short connect J1 plug on external calling serial board, and it will show the setting floor address. Press the up button on the calling board, the floor address will add 1, and press the down button on the calling board, the floor address will decrease 1. If the button is not pressed in the continuous 3S, the current data is stored as the floor address on the display board. If the floor data flashes twince indicating that the storage is successful, and the button is invalid during the flashing period.



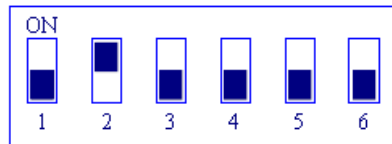
The display board address in the car must be set to 0, otherwise it will not display overload!



The external calling floor address is the physical floor number (take the levelling board as the standard), increasing from bottom to top, but is not related to the displayed content of the actual floor!

4.3 Dip switch external calling setting method:

Set corresponding binary encoding mode "ON" and "OFF" for the address dip switch on external calling serial board on each floor according to the elevator electrical principle diagram, and the dip switch from 1 to 6 are the most low to the most high position:



The figure above is 010000, the decimal conversion is 2, representing the second floor address of the absolute floor. The serial display will flicker for three times "2", it represents the absolute address code.

The address of the calling board has been set up before delivery, and it can be installed according to the position of the floor on the panel!

Chapter 5 Elevator Commissioning

Notice

Before the trial operation, it is necessary to confirm that the shaft and the related devices are fully installed and meet the requirements of the rated speed operation of the elevator.

1. Automatic Leveling Function Operation

- 1.1 Operate elevator emergency to the middle position between two floors;
- 1.2 Reset the emergency electric operation switch, and the elevator is in the state of automatic operation. At this time, the elevator will run to the leveling position of the nearest floor and stop, and then automatically open the door.



Self-leveling height can be adjusted by the FD-05 parameter (when it exceeds levelling position, the parameter will reduce; when it does not reach the level position, the parameter will increase)!

2. Single Floor Operation

- 2.1 Put the elevator in normal running condition;
- 2.2 Enter the single floor operation instructions;
- 2.3 The elevator will run at the single floor speed. Observe the operation of the elevator, and confirm that the elevator can start, accelerate, slow down, level and stop;
- 2.4 During the elevator operation, if abnormal phenomenon is found, must immediately press the emergency stop switch, so that the elevator is forced to stop. Carefully check the elevator. After troubleshooting, the elevator can operate again.

3. Multi-floor Operation

- 3.1 Enter the multi-floor operation instructions;
- 3.2 The elevator will run at the multi-floor speed. Observe the operation of the elevator, and confirm that the elevator can start, accelerate, slow down, level and

stop;

- 3.3 During the elevator operation, if abnormal phenomenon is found, must immediately press the emergency stop switch, so that the elevator is forced to stop. Carefully check the elevator. After troubleshooting, the elevator can operate again.

4. Confirmation of Door Opening and Closing Operation

4.1 Check the door opening and door closing button

- 4.2 Under the inspection door zone position, press the door opening button, and the door opens; press the door closing button, and the door closes;

- 4.3 In the automatic state, the elevator automatically opens the door after stopping and levelling, and closes automatically after waiting for a period of time. The opening/closing operation can be realized through door opening / closing button in car;

- 4.4 Under the driver's condition, the elevator automatically opens the door after stopping and levelling, and closes after continuously pressing the door closing button. Loosen the door closing button before the door is not closed completely, then the elevator opens the door;

- 4.5 Confirm the front and rear door action are in normal state before double doors open (elevator opposite door usually opens at same time when arriving at the position);



1. The door opening function of the elevator is invalid if not at the door zone position;

2. Under the inspection mode, before the elevator does not start operation, when pressing the up / down running button, the system will not output the door closing signal. The door closing signal inching operation must be done through the small key on the main board. The specific operation method is as follows:

Enter into the F-E menu with the keys PRG, UP, SET on the small keyboard, and the digital tube will show 1-1. At this time, UP and SET keys represent door inching opening and door inching

closing respectively. Press the PRG key to quit the operation mode.

5. Confirmation of Instruction Registration and Self-protection Function

5.1 The instruction registration must be in elevator's normal state;

5.2 When the elevator is in the normal collective selection state, press all instructions and call buttons one by one, corresponding indicator lights except this elevator will be on and hold on. If they are not on, check whether the corresponding wiring is correct or not;

5.3 When the elevator is in the driver state, press all the instructions in car, the corresponding indicators except this floor of the elevator will be on and hold on.

If they are not on, check whether the corresponding wiring is correct or not; Press all hall call instructions, the corresponding indicators except this floor of the elevator will be on and hold on, and the corresponding indicators in the car will shine. If there is no corresponding phenomenon, check whether the corresponding wiring is correct or not.

6. Commissioning of Elevator Duplex Control Function

6.1 Confirm the duplex communication line is connected correctly according to the requirements of electrical principle diagram:

6.2 Determine the duplex control function parameters of the main board:

Function code	Explanation	Setting scope	Setting value
F6-07	Group control number	1-8	2
F6-08	Elevator number	1-8	Set main elevator to 1 Set auxiliary elevator to 2
F6-09	Program selection	Bit3:CAN2 duplex communication function	8 (1000)

6.3 Duplex external calling address setting:

Two sets of elevators are in duplex. When the two elevators stop at the same floor, and the related parameters and the hall calling address should be the same as the simplex control elevator; When the two elevators do not stop at the same floor, the elevator hall calling address setting will be set different from simplex control elevator. The following is a typical example. The practical use shall be flexible according to the sample.

Example: Suppose there are two elevators I duplex control, one stops at B1 floor, 1 floor, 2 floor, 3 floor, and the other stops at 1 floor, 3 floor, 4 floor, then the two elevators' related parameters and hall call address are as follows:

		Number 1 elevator		Number 2 elevator	
Group control number (F6-07)		2		2	
Elevator number (F6-08)		1		2	
User floor	Physical floor	Hall calling address	Hall calling display	Hall calling address	Hall calling display
B1	1	1	FE-01=1101		
1	2	2	FE-02=1901	2	FE-02=1901
2	3	3	FE-03=1902	The elevator shall not stop at this floor, but this floor must be provided with leveling board	FE-03=1902
3	4	4	FE-04=1903	4	FE-04=1903
4	5			5	FE-05=1904
The lowest floor (F6-01)		1		2	
The highest floor (F6-00)		4		5	
1-16 service floor (F6-05)		65535 (Select according to BI0 ~ BI16)		65531 (physical floor floor 3 does not stop) (Select according to BI0 ~ BI16)	
17-32 service floor (F6-06)		65535 (Select according to BI0 ~ BI16)		65535 (Select according to BI0 ~ BI16)	

7. Confirmation of Elevator Balance Coefficient

7.1 Technical requirements: the balance coefficient is 40% ~ 50% (recommend 45%);

7.2 The car carries out up and down operation with 45% of the rated load. Measure the motor stator current when the car runs to the middle of the journey, and increase or reduce the counterweight filler until the elevator's current value for up and down is equal, and the car load shall be within 40% ~50%;

7.3 Re-test: The car operates up and down at no-load and 25%, 75%, 100% of rated load. When the car and counterweight run to the same level position, record the current value of each load, and draw the load current curve, and determine the balance coefficient according to the intersection of up and down operation load - current curve.

8. Confirmation of Elevator Automatic Evacuation Device Function (no need to confirm if it is not provided)

8.1 Confirm the correct connection wiring according to the requirements of electrical principle diagram;

8.2 Determine the control main board functional parameters:

Function code	Explanation	Setting scope	Setting value
F8-10	Selection of rescue method	1-2	1
F5-20	Rescue operation signal		27

9. Elevator opposite door commissioning (no need to set if there is no this function)

9.1 **Opposite door control mode and the corresponding parameter settings description:**

Control mode	Parameter setting		Function description	Method of use
Mode 1	FE-33=32806; Fb-00=2;	FC-04=0	Opposite door control at the same time: regardless of the hall calling or the car calling arrives, the front and rear door simultaneous act;	Front door hall calling address settings (1~15), rear door hall calling address settings (floor number +16);
Mode 2		FC-04=1	The hall calling is independent, and the car calling is consistent: when the hall calling arrives, the door opens correspondingly. When the car calling arrives, the front and the rear door simultaneously act;	
Mode 3		FC-04=3	The hall calling is independent and the car calling is independent: The hall calling and the car calling correspond to door opening;	

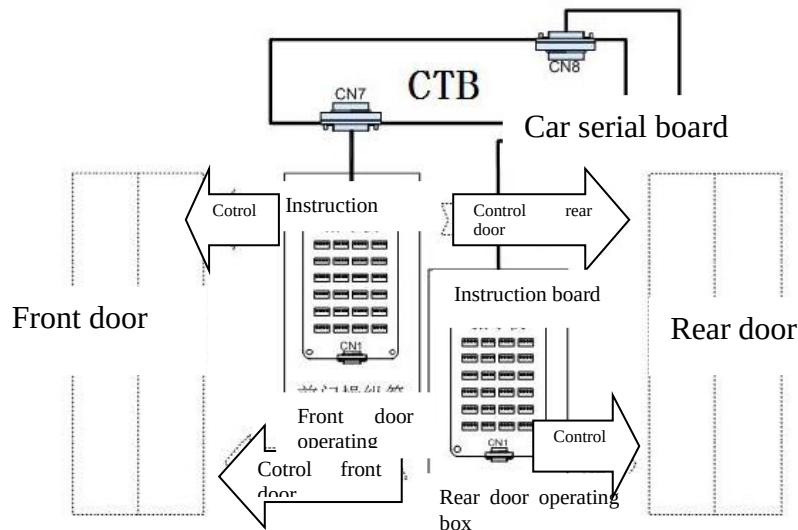
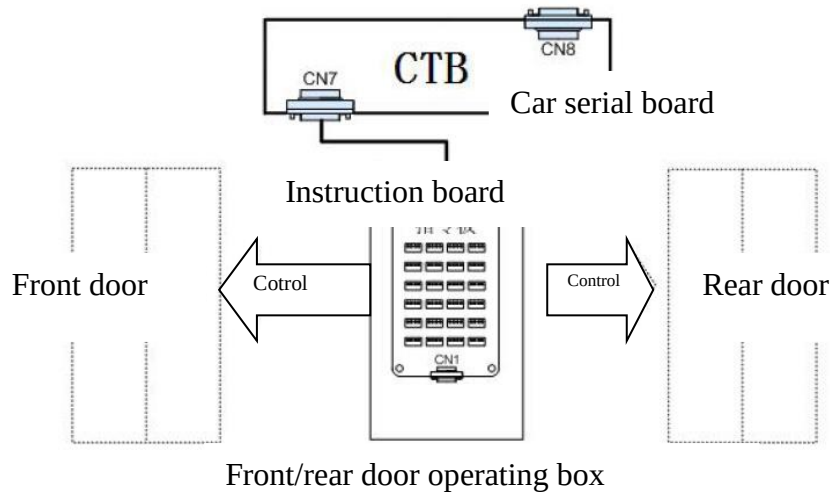


Elevator opposite door service floor shall be set separately through FB-02 ~ FB-05 parameter group!

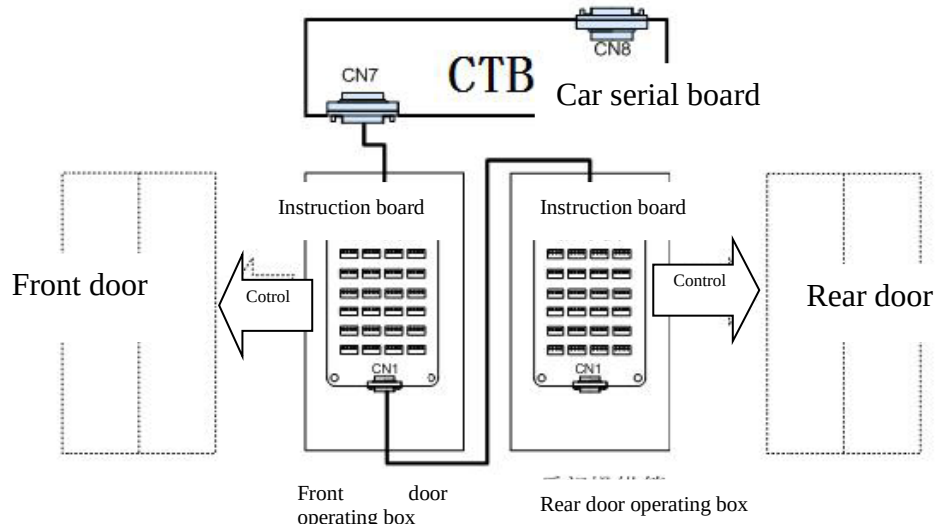
In the process of fire control and elevator locking, the function of opposite shall not be independently controlled!

9.2 Car instruction board wiring description of the opposite door function:

- Apply to opposite door function control mode 1 and control mode 2:



- Apply to opposite door function control mode 3:



10. Instructions of Asynchronous Motor Auxiliary Brake (no need to set if there is no this function)

- 10.1 Confirm that the auxiliary brake has been installed and adjusted in place.
(installation method is shown in the documents of auxiliary brake);
- 10.2 When the additional brake is not energized, confirm the auxiliary brake's action detection switch is in the state of release (no matter the auxiliary brake action bolt is in any position), and confirm X22 on the main control board has no input signal;
- 10.3 When the auxiliary brake is energized, confirm whether the auxiliary brake coil is energized, and that the brake action detection switch is in action state, and that the X22 on the main control board has input signal;
- 10.4 Confirm whether the function parameters have been open or not:

Function code	Name	Factory setting
A1-05	Function selection	8 (1000)
F5-16	X16 function selection	89
F5-17	X17 function selection	90
F5-22	X 22 function selection	129
F5-29	Y4 function selection	54
F6-55	Function selection	512

11. Door Lock Bypass Function

A door lock bypass module is provided in the control cabinet to bypass the landing door or the car door lock circuit. When the elevator is in normal condition, bypass module should be inserted in the plug-in of normal mode, and the corresponding LED indicator of normal module will be on. When the elevator needs bypass lock maintenance operation, this module can be used to bypass any lock circuit of the car door or landing door lock circuit, and at the same time, the corresponding door lock bypass LED status indicator will be on, and automatically enter the inspection or emergency electric operation mode.



- 1. The door lock bypass operation can only passby any one group of the lock circuit of the landing door or car door!**
- 2. In the door lock bypass operation, the system automatically enters the emergency electric /inspection operation mode!**

3. In elevator door lock bypass operation, there still needs to collect and monitor the door closing in place signal!

12. Elevator Car Unintended Movement Detection Function Test (no need to set if no this function)

12.1 Confirm that the elevator re-levelling function is open, and that the related parameters are set correctly:

12.2 Safety tips before unintended movement detection function test:

 Safety tips

- Before the fault simulation, it must be confirmed that the elevator has been adjusted and meets the requirements of acceptance. Safety protection work shall have been done. Be familiar with the following operating rules!
- In test, it shall be ensured that the car is empty and is without personnel. After the test is completed, all faults and action parts shall be reset (including auxiliary brake)!

12.3 Fault simulation test operation steps of unintended movement detection function:

- a) The elevator in the normal mode runs to levelling position, and at the same time it is necessary to ensure the safety of the elevator and the closing of the door lock circuit;
- b) When the machine room is switched into emergency electric mode, it is necessary to ensure the safety of the elevator and the closing of the door lock circuit (otherwise it cannot start the car unintended movement simulation test function);
- c) Set the "F-8" function code to "7" through the the small keyboard on the mainboard, and the system enters the simulation test mode automatically, and the mainboard displays "E88" at this time;
- d) Pull out the UCMP-T test plug-in components in the control cabinet and simulate the disconnection of the door lock circuit;
- e) If synchronous motor is used, the operator shall press upward+ confirmation button or downward+ cofirmation button in the inspection box, and the door closing signal is output from the main board. The short circuit door lock is output by safety circuit of SMZ safety module, and the elevator runs in inspection mode. When the elevator is break away from the door zone, the SMZ module

immediately cancels the door closing function, and the system reports "E65" fault, and the elevator stops running;

- f) If the asynchronous motor is used, the operator shall open the working brake by hand, and the elevator slides. When the car is break away from door zone, the system immediately reports "E65" fault, and the elevator stops running;
- g) If synchronous motor is used, use the working brake to stop the car movement. If asynchronous machine is used, the independent auxiliary brake shall be used to stop the car movement. After stopping the car, measure whether the car moving distance meets the requirements of standard.

★ **In order to ensure safety, the elevator shall be operated to the floor beneath the top floor and carry out simulation test!**

★ **In order to ensure safety, the actual lock circuit must be in closed state in the whole simulation test process!**

★ **"E65" fault will not automatically reset and can only be reset by setting "F-2" to "1" by the small keyboard on the mainboard in inspection or emergency electric mode!**

13. Machine Braking Force Monitoring Function Test (no need to set if no this function)

13.1 The brake force function test is only suitable for gearless traction machine;

13.2 Safety tips before brake force monitoring function test:

 Safety tips

- Before the fault simulation, it must be confirmed that the elevator has been adjusted and meets the requirements of acceptance. Safety protection work shall have been done. Be familiar with the following operating rules!
- In test, make sure that the car is empty and is without personnel. After the test is completed, all faults and action parts shall be reset (including auxiliary brake)!

13.3 Brake force monitoring function test procedures:

- a) The elevator runs in the normal mode to the leveling position of top floor (ensure that the car is empty and without personnel), and at the same time it is necessary to ensure the safety of the elevator and the closing of the door lock circuit;
- b) When the machine room is switched into emergency electric mode, it is necessary to ensure the safety of the elevator and the connection of the door lock circuit

(otherwise it cannot start the brake force simulation test function);

- c) Set the "F-8" function code to "8" through the small keyboard on the main board, and the system enters the braking force simulation test mode automatically, and the mainboard displays "E88" at this time;
- d) The main controller outputs the star-delta contactor and the operation contactor action signal, and at this time, the KMYF star-delta contactor and the KMY operation contactor pull in;
- e) The motor drive command is effective, and the driving chip outputs to the traction machine to do braking force test according to the preset torque current, and "E88" has been shown in the process of detection;
- f) After the braking force detection is finished, the control board automatically closes star-delta contactor and operation contactor. If the detection is normal, "E88" on the main board is changed to flicker from static state. Press the "PRG" key on the mainboard to exit the brake force test mode; if the braking force detection is not normal, the system reports "E66" fault.

- ★ **In order to ensure safety, the elevator shall be operated to the top floor and carry out brake force simulation test!**
- ★ **In order to ensure safety, the actual lock circuit must be in closed state in the whole simulation test process state, otherwise automatically exists test mode!**
- ★ **"E66" fault cannot reset automatically. The cause of the fault must be found and then carry out brake force test again, otherwise the system has always been in a breakdown state!**

Chapter 6 Adjustment of elevator leveling and comfort

1. Elevator Leveling Adjustment

Operation direction	Car position	Adjustment method
Upward leveling	Low	Increase F4-00
	High	Decrease F4-00
Downward leveling	Low	Decrease F4-00
	High	Increase F4-00

2. Elevator Starting Comfort Adjustment

Elevator starting comfort	Adjustment method
Start reverse slide	Appropriately increase F2-11
Start too fiercely	Appropriately reduce F2-11 and reduce F2-00

3. Elevator Operation Comfort Adjustment

Elevator operation comfort at high speed	Adjustment method
When the elevator runs at high speed, the car has a feeling of trembling up and down	Appropriately increase F2-03 or reduce F2-04
When the elevator runs at high speed, the car has a feeling of vibration upward	Appropriately reduce F2-03 or increase F2-04



Tips:

- 1. Adjust the proportional gain F2-03 at first during the adjustment of comfort!**
- 2. The improper F2-03 and F2-04 parameters adjustment may lead to excessive overspeed and even over-voltage!**
- 3. Too small F2-00 parameter setting will lead to elevator abnormal leveling!**

Appendix 1 Instructions for commissioning tools

1. Hand-held Operator Diagram

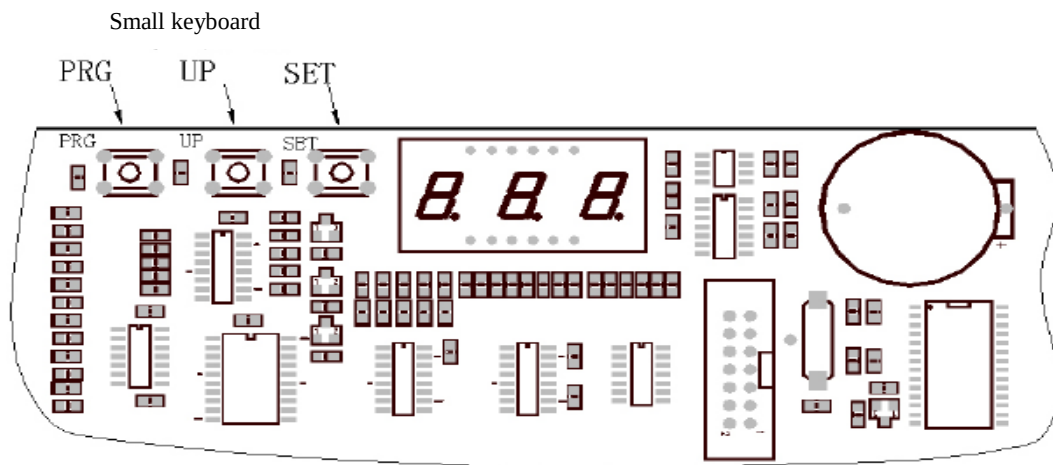


2 Button Function Instruction of Hand-held Operation Panel Keyboard

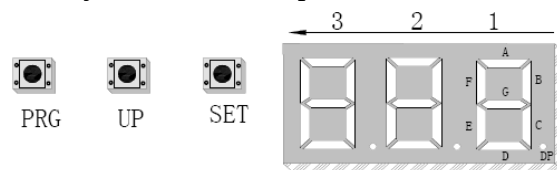
Button	Name	Function description
	Left key	Perform the display function of the screen lower left corner
	Right key	Perform the display function of the screen lower right corner
	Knob confirmation key	Can increase or decrease number by turning to set parameters and choose the menu
	Running key	In the keyboard operation control mode, used to start the operation
	Stop key	In the keyboard operation control mode, used to stop operation; in the fault alarm state, it can be used for reset operation
	PRG key	Return to previous menu

3. Function and Instructions of small keyboard

3.1 Small keyboard appearance:



3.2 Small keyboard key function description:



Key name	Function description
PRG key	Function menu key: In any state, press the PRG key, it will display the current function group menu number which can be changed through the UP key
UP key	Direction button: - In the function group menu, menu cycle increasing is realized by UP key, and the current main controller has 13 functional group menu, that is, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C,; - Input data through UP key;
SET key	Confirmation key: - In the function group menu, press the SET button to enter the data menu of the function group; - The data under the function menu can be saved by pressing the SET key. After saving the data, the operation panel tacitly enters the

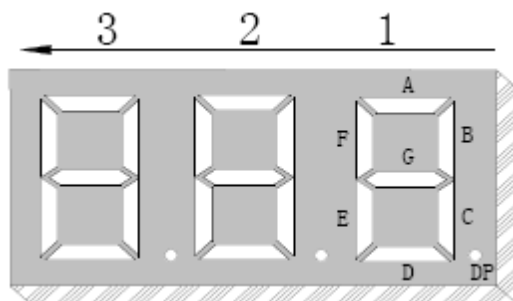
	menu display of F-0;
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3.3 Small keyboard display menu classification:

Function menu	Menu contents	Detailed description
F-0	Floor and operation direction information	Power up default is F-0 data menu display, the last two digits of the 3 digital tube display the current elevator floor, and the first digital tube displays direction. When the elevator stops, the first digital tube does not display. When the elevator runs upward or downward, the digital tube indicates the upward or downward direction. In the case of system failure, the digital tube automatically switches to the fault code flashing, and if the fault disappears automatically, the digital tube automatically enters the F-0 state display;
F-1	Floor instruction command input	In the automatic mode, enter the F-1 menu, the digital tube displays the minimum elevator floor, and the data setting can be changed by using the UP key with the range from the minimum floor to the maximum floor. Press SET to save the data after setting, and the elevator runs to the specified destination floor;
F-2	Fault reset and code display	Enter the menu, and the digital display "0". The range "1~2" can be set. " 1 " represents system fault reset command, and press the SET key to clear the current system fault;" 2 " represents system fault code, and press SET and the digital tube will cyclically display 11 fault code;
F-3	Time display	After entering the menu, the digital tube automatically displays the time, such as "2017-01-18-10-00";
F-5	Running times display	After entering the menu, the digital tube cyclically display the running times;
F-6	Set elevator maximum floor	Set elevator maximum floor, function equivalent to F6-00; (Note: can only set but cannot see it; enter F-6, and the initial value is 2)
F-7	Shaft floor height self-learning	Enter the menu in the inspection mode, set the parameter to "1", and the system automatically enters the floor height self-learning, and the

		function is equivalent to "3" of F1-11;
F-8	Unintended movement simulation Test function selection	Set to "7" to simulate unintended movement test (SMZ security module must be equipped, and the re-levelling function shall be open);
		Set to "8" to simulate the braking force test, only applicable to gearless machine;
F-A	Traction machine self-learning	After entering the menu under inspection mode, set the parameter to "1", and the system enters the machine self-learning state, and the function is equivalent to F1-11;
F-B	Car serial board status display	See details in description;
F-C	Elevator running direction change	Change the direction of elevator operation, function equivalent to F2-10: 0: Running in the same direction; 1: Running in the opposite direction;
F-E	Door opening and closing control in inspection mode	Under inspection mode, enter menu display "1-1", and at this time press "UP" key to do inching door opening, and press the "SET" key to do inching door closing;

F-B: Car serial board status display



Digital tube Number	Digital tube segment mark	Definition of signal	The meaning of "on" in digital tube section
1	A	The front door light curtain signal	When input the signal of front door light curtain, the section A of No.1 digital tube is on
	B	The rear door light curtain signal	When input the signal of rear door light curtain, the section B of No.1 digital tube is on
	C	Door opening in place 1	When input the signal of front door open in place, the section C of No.1 digital tube is on

	D	Door opening in place 2	When input the signal of rear door open in place, the section D of the No. 1 digital tube is on
	E	Door closing in place 1	When input the signal of front door close in place, the section E of No.1 digital tube is on
	F	Door closing in place 2	When input the signal of rear door close in place, the section F of No.1 digital tube is on
	G	Full load signal	When input the signal of full load, the section G of No.1 digital tube is on
	DP	Overload signal	When input the signal of overload, the section DP of the first digital tube is on
2	A	Light load signal	When input the signal of light load, the section A of No. 2 digital tube is on
3	A	Front door opening output	When output the signal of front door open, the section A of No.3 digital tube is on
	B	Front door closing output	When output the signal of front door close, the section B of No.3 digital tube is on
	C	Front door forced closing output	When output the signal of front door forced closing, the section C of No.3 digital tube is on
	D	Rear door opening output	When input the signal of rear door open, the section D of No.3 digital tube is on
	E	Rear door closing output	When input the signal of rear door close, the section E of No.3 digital tube is on
	F	Rear door forced closing output	When output the signal of rear door forced closing, the section F of No.3 digital tube is on
	G	Upward arrival gong output	When output the signal of upward arrival gong, the section G of No.3 digital tube is on
	DP	Downward arrival gong output	When output the signal of downward arrival gong, the section DP of No.3 digital tube is on

Appendix 2 Controller Function Parameters Description

Function code	Name	Setting scope	Factory setting	Note
Group F0 Basic parameters				
F0-00	Control method	0: Open loop vector 1: Closed loop vector	1	
F0-01	Command source selection	0: Operation panel control 1: Distance control	1	
F0-03	Elevator's maximum running speed	0.250~F0-04	According to elevator's actual running speed	
F0-04	Elevator's rated speed	0.250~4.000m/s	According to elevator's rated speed	
F0-05	Elevator's rated load	300~9999 kg	According to elevator's actual rated load	
F0-06	Maximum frequency	20.00Hz~99.00Hz	According to actual motor frequency	
F0-07	Carrier frequency	0.5~16.0kHz	6kHz	
Group F1 Motor parameters				
F1-00	Encoder type selection	0: SIN/COS 2: ABZ encoder	0 (synchronous motor) 2 (asynchronous motor)	
F1-01	Rated power	1.1~75.0Kw	According to actual motor rated power	
F1-02	Rated voltage	0~440V	According to actual motor rated voltage	
F1-03	Rated current	0.00~655.00A	According to actual motor rated current	
F1-04	Rated frequency	0.00~99.00Hz	According to actual motor rated frequency	
F1-05	Rated rotate speed	0~3000rpm	According to actual motor rated rotate speed	
F1-06	Initial angle of synchronous motor	0~359.9°	Initial angle of synchronous motor	Automatic record
F1-07	Synchronous motor power off angle	0~359.9°	Synchronous motor power off angle	
F1-08	Synchronous motor wiring method	0~15	0	
F1-09	Current filter coefficient of synchronous motor	0~40	0	
F1-10	Encoder check selection	0~65535	0	
F1-11	Self-learning selection	0: No operation 1: Motor with load tuning 2: Motor no-load tuning 3: self-learning of shaft floor height	0	
F1-12	Number of pulses per turn of encoder	0~10000	2048	
F1-14	Asynchronous motor's stator resistance	0~30		
F1-15	Asynchronous motor's rotor resistance	0~30		
F1-16	Leakage inductance of asynchronous machine	0~300		

Function code	Name	Setting scope	Factory setting	Note
F1-17	Mutual inductance reactance of asynchronous motor	0~3000		
F1-18	Asynchronous motor's no-load current	0.010~300		
F1-19	Q axis inductance	0~650		
F1-20	D axis inductance	0~650		
F1-25	Motor type	0: Asynchronous motor 1: Synchronous motor	1	
Group F2 Vector control parameters				
F2-00	Speed loop proportional gain 1	0~100	40	
F2-01	Speed loop integration time 1	0.01~10.00s	0.60s	
F2-02	Switching frequency 1	0.00~F2-05	2.00Hz	
F2-03	Speed loop proportional gain 2	0~100	35	
F2-04	Speed loop integration time 2	0.01~10.00s	0.80s	
F2-05	Switching frequency 2	F2-02~F0-06	5.00Hz	
F2-06	Current loop proportional gain	10~500	220	Self-learning and self-generating
F2-07	Current loop integration gain	10~500	80	
F2-08	Torque upper limit	0~200	180%	
F2-10	Elevator running direction	0: Running in the same direction; 1: Running in the opposite direction;	0	
F2-11	Zero servo current coefficient	0.20%~50.0%	15.0 (Pre-tuning mode 2)	F8-01=2 mode
F2-12	Zero servo speed loop KP	0.00~1.00	0.50 (Pre-tuning mode 2)	
F2-13	Zero servo speed loop TI	0.00~2.00	0.60 (Pre-tuning mode 2)	
F2-38	Detection torque amplitude	0~150%	According to different motors' calculation value	
Group F3 Operation control parameters				
F3-02	Acceleration	0.200~2.000m/s ²	0.600m/s ²	
F3-03	Inflection point acceleration time 1	0.300~4.000s	3.000s	
F3-04	Inflection point acceleration time 2	0.300~4.000s	3.000s	
F3-05	Deceleration	0.200~2.000m/s ²	0.600m/s ²	
F3-06	Inflection point deceleration time 1	0.300~4.000s	3.000s	
F3-07	Inflection point deceleration time 2	0.300~4.000s	3.000s	
F3-08	Special deceleration	0.500~2.000m/s ²	0.900m/s ²	
F3-09	Parking distance margin	0~90.0mm	0.0mm	
F3-11	low operating speed	0.100~0.630 m/s	0.250m/s	
Group F4 floor parameter				

Function code	Name	Setting scope	Factory setting	Note
F4-00	Leveling adjustment	0~60mm	30mm	
F4-01	Current floor	F6-01~F6-00	1	
F4-02	High position of elevator current position	0~65535		
F4-03	Low position of elevator current position	0~65535		
Group 5 terminal function parameters				
F5-01	X1 function selection		01	
F5-02	X2 function selection		03	
F5-03	X3 function selection		02	
F5-04	X4 function selection		00 (without down triple reduction) 53 (with down triple reduction)	
F5-05	X5 function selection		05 (with KMM) 00 (without KMM)	
F5-06	X6 function selection		38	
F5-07	X7 function selection		39	
F5-08	X8 function selection		30 (with KMYF) 00 (without KMYF)	
F5-09	X9 function selection		40	
F5-10	X10 function selection		09	
F5-11	X11 function selection		10	
F5-12	X12 function selection		00 (without pit inspection function) 120 (with pit inspection function)	
F5-13	X13 function selection		00 (without overload switch/narmally colsed) 46 (with overload switch/narmally colsed)	
F5-14	X14 function selection		48	
F5-15	X15 function selection		49	
F5-16	X16 function selection		00 (speed≤1.75m/s) 50 (speed>1.75m/s)	Synchronous motor
			00 (without UCMP) 89 (with UCMP)	Asynchronous motor
F5-17	X17 function selection		00 (speed≤1.75m/s) 51 (speed>1.75m/s)	Synchronous motor
			00 (without UCMP) 90 (with UCMP)	Asynchronous motor
F5-18	X18 function selection		00 (witout brake switch) 58 (with brake switch, normally closed)	
F5-19	X19 function selection		00 (without brake switch) 110 (with brake switch, normally closed)	
F5-20	X20 function selection		00 (without automatic evacuation function)	

Function code	Name	Setting scope	Factory setting	Note
			27 (with automatic evacuation function)	
F5-21	X21 function selection		11	
F5-22	X 22 function selection		00 (without up triple reduction) 52 (with up triple triple reduction)	Synchronous motor,with UCMP
			64 (external fault)	Without machine room
			00 (without UCMP) 129 (with UCMP)	Asynchronous motor
F5-23	X23 function selection		00 (without re-leveling function) 22 (with re-leveling function)	
F5-24	X24 function selection		00 (without motor thermistor switch) 99 (with motor thermistor switch)	
F5-25	Car roof panel input type selection	0～511	380 (101111100)	With door endswitch
			320 (101000000)	Without door endswitch
F5-26	Y1 function selection		1	
F5-27	Y2 function selection		2	
F5-28	Y3 function selection		16	
F5-29	Y4 function selection		00 (without KMYF) 12 (with KMYF)	Synchronous motor
			00 (Without UCMP) 54 (With UCMP)	Asynchronous motor
F5-30	Y5 function selection		04	
F5-31	Y6 function selection		00 (without re-leveling function) 03 (with re-leveling function)	
F5-36	Weighing input selection		1	Weighing of the car top
			0	Weighing of machine room
F5-37	X25 function selection		4	
F5-38	X26 function selection		5	
F5-39	X27 function selection		5	Without door circuit short checkin
			6	With door circuit short checkin
Group F6 Elevator basic parameters				
F6-00	Maximum floor of the elevator	1～40	According to the actual floor of the elevator	
F6-01	Minimum floor of the elevator	1～F6-00	1	
F6-02	Parking floor	F6-01～F6-00	1	
F6-03	Fire base station	F6-01～F6-00	1	
F6-04	Elevator locking floor	F6-01～F6-00	1	
F6-07	Group control	1～8	1	

Function code	Name	Setting scope	Factory setting	Note
	number			
F6-08	Elevator number	1～8	1	
F6-10	levelling inductor delay	10～50ms	14ms	
F6-11	Elevator function selection		8448（10000100000000）	
F6-40	Elevator function selection		2 （10）	
F6-41	Elevator function selection		4610 （1001000000010）	
F6-42	Elevator function selection		8 （1000）	
F6-43	Elevator function selection		130 （10000010）	
F6-44	Elevator function selection		16456（100000001001000）	
F6-45	Elevator function selection		1024 （10000000000）	
F6-55	Elevator function selection		00（Without UCMP） 512（With UCMP）	Asynchronous motor
Group F8 Enhancement function parameters				
F8-01	Pre-torque selection	0: invalid pre-torque 1: weighing pre-torque compensation 2: pre-torque automatic compensation	2	
F8-19	Elevator function selection		768（1100000000）	Without UCMP
			0	With UCMP
Group F9 Time parameters				
F9-00	Time of returning to base floor in idle time	0～240min	10 min (return to base floor after 10 min) 0 min (do not return to base floor)	
F9-01	Fan and lighting closing time	0～240min	5min	
F9-02	High speed operation protection time	0～45s (no effect under 3s)	45s	
F9-11	Running times at high position	0～9999	0	
F9-12	Running times at low position	0～9999	0	
Group FB Door machine function parameters				
FB-00	Door machine quantity	1～2	1	
FB-02	Front door service floor (1-16)	0～65535	65535（1111111111111111）	
FB-03	Front door service floorr (17-32)	0～65535		
FB-04	Rear door service floor (1-16)	0～65535		
FB-05	Rear door service floor (17- 32)	0～65535		
FB-08	Door closing time protection	5～99	15	
FB-09	Door closing/opening times	0～20	0	

Function code	Name	Setting scope	Factory setting	Note
FB-18	Front door service floor (33- 48)	0~65535	65535 (1111111111111111)	
FB-19	Rear door service floor (33- 48)	0~65535		
FB-11	Hall calling door opening time	1~1000	5s	
FB-12	Car calling door opening time	1~1000	3s	
Group FC Protection function parameters				
FC-00	Protection detection selection		2 (10)	With door endswitch
			514 (1000000010)	Without door endswitch
FC-01	Protection function selection		65 (1000001)	
Group FE Elevator function setting parameters				
FE-00	Collective selective method	Full collective selective selective	0	
FE-01	Floor 1 corresponding display	0000~1999 High two bit represents ten digit display code; low two bit represents single digit display code; the display code is as follows: 00: display "0" 01: display "1" 02: display "2" 03: display "3" 04: display "4" 05: display "5" 06: display "6" 07: display "7" 08: display "8" 09: display "9" 10: display "A" 11: display "B" 12: display "G" 13: display "H" 14: display "L" 15: display "M" 16: display "P" 17: display "R" 18: display "-" 19: without display 20: display "12" 21: display "13" 22: display "23" 23: display "C" 24: display "D" 25: display "E" 26: display "F" 27: display "I" 28: display "J" 29: display "K" 30: display "N" 31: display "O" 32: display "Q" 33: display "S"	1901	
FE-02	Floor 2 corresponding display		1902	
FE-03	Floor 3 corresponding display		1903	
FE-04	Floor 4 corresponding display		1904	
FE-05	Floor 5 corresponding display		1905	
FE-06	Floor 6 corresponding display		1906	
FE-07	Floor 7 corresponding display		1907	
FE-08	Floor 8 corresponding display		1908	
FE-09	Floor 9 corresponding display		1909	
FE-10	Floor 10 corresponding display		0100	
FE-11	Floor 11 corresponding display		0101	
FE-12	Floor 12 corresponding display		0102	
FE-13	Floor 13 corresponding display		0103	
FE-14	Floor 14 corresponding display		0104	

Function code	Name	Setting scope	Factory setting	Note	
FE-15	Floor 15 corresponding display	34: display "T" 35: display "U" 36: display "V" 37: display "W" 38: display "X" 39: display "Y" 40: display "Z" 41: display "15" 42: display "17" 43: display "19"	0105		
FE-16	Floor 16 corresponding display		0106		
FE-17	Floor 17 corresponding display		0107		
FE-18	Floor 18 corresponding display		0108		
FE-19	Floor 19 corresponding display		0109		
FE-20	Floor 20 corresponding display		0200		
FE-21	Floor 21 corresponding display		0201		
FE-22	Floor 22 corresponding display		0202		
FE-23	Floor 23 corresponding display		0203		
FE-24	Floor 24 corresponding display		0204		
FE-25	Floor 25 corresponding display		0205		
FE-26	Floor 26 corresponding display		0206		
FE-27	Floor 27 corresponding display		0207		
FE-28	Floor 28 corresponding display		0208		
FE-29	Floor 29 corresponding display		0209		
FE-30	Floor 30 corresponding display		0300		
FE-31	Floor 31 corresponding display		0301		
FE-35 ~ FE-43	Floor 32~40 corresponding display				
FE-32	Function setting selection 1			34832 (without re-leveling function)(1000100000010000)	
				34836 (with re-leveling function) (10001000000010100)	

Function code	Name	Setting scope	Factory setting	Note
FE-33	Function setting selection 2		38 (without opposite door function) (100110)	
			32806 (with opposite door function) (1000000000100110)	
Group A1 Elevator enhancement function setting parameters				
A1-05	Function setting selection		0	Synchronous motor
			8 (1000)	Asynchronous motor, with UCMP
A1-08	Function setting selection		8 (1000)	
A1-09	Return level floor	0 ∼ F6-00	0	
A1-10	Function setting selection		33847 (1000010000110111)	
A1-11	Function setting selection		10	
A1-16	Problem floor monitoring		0000	Monitoring function
A1-26	Floor display function selection	0∼3	0 (FE parameters for all display); 1 (FE parameters for car and parallel control for the hall); 2 (adopt parallel control for the car and FE parameters for the hall); 3 (parallel control for all display);	



The parameters of group A1 in the controller are enhancement function Parameters, only applicable to F04.05 and above procedures ,and cannot be arbitrarily modified without any consent so as to avoid other faults!

Appendix 3 Fault Information

Operation panel display	Small keyboard display	Fault description	Fault reason	Processing method
Err01	E01	Inverter unit protect	1. Main circuit output grounding or short circuit; 2. Traction machine connection is too long; 3. Overheating working environment; 4. Loose wiring inside the controller;	1. Exclude wiring and other external problems; 2. Check whether the air duct and fan are normal; 3. Please contact the technical department;
Err02	E02	Accelerating over-current	1. Main circuit output grounding or short circuit; 2. Whether the motor has been tuned; 3. Too much load; 4. Encoder signal is incorrect;	1. Check whether the inverter output side and the operation contactor are working properly; 2. Check whether the motor power line connection is reliable, whether the surface is damaged, and whether there is short circuit to the ground;
Err03	E03	Deceleration over-current	1. Main circuit output grounding or short circuit; 2. Whether the motor has been tuned; 3. Too much load; 4. The deceleration curve is too steep; 5. Encoder signal is incorrect;	3. Check whether the motor has short circuit or grounding; 4. Check whether the motor parameters are consistent with the nameplate; 5. Do again self-learning of motor parameters;
Err04	E04	Constant speed over-current	1. Main circuit output grounding or short circuit; 2. Whether the motor has been tuned; 3. Too much load; 4. Rotary encoder has great interference;	6. Check whether the brake work is normal; 7. Check whether the encoder related wiring is correct and reliable; 8. Check whether the acceleration and deceleration are too large;
Err05	E05	Acceleration over-voltage	1. Input voltage is too high; 2. The elevator is seriously pulled up; 3. The braking resistance is too large or the brake resistance is abnormal; 4. The acceleration curve is too steep;	1. Adjust the input voltage, and observe whether the bus voltage is normal, and whether it rises too fast in operation; 2. Check the balance coefficient of elevator;
Err06	E06	Deceleration over-voltage	1. Input voltage is too high; 2. The braking resistance is too large or the brake resistance is abnormal; 3. The deceleration curve is too deep;	3. Check whether the selected brake resistor is suitable; 4. Check whether the brake resistance wiring is broken, whether there is grounding phenomenon, and whether the wiring is reliable;
Err07	E07	Constant speed over-voltage	1. Input voltage is too high 2. The braking resistance is too large or the brake resistance is abnormal	
Err08	E08	Maintenance warning	Within the setting time, the elevator does not receive power-off maintenance	In the specified time, the elevator needs maintenance under power off condition to ensure the safety of the elevator
Err09	E09	Under-voltage fault	1. There is sudden power cut for the input power; 2. The input voltage is too low; 3. The drive control panel is abnormal;	1. Exclude external power problems; check whether there is a disconnection of the power supply in operation; 2. Check whether all power input lines and connection pile head are reliable;
Err 10	E10	System overload	1. The brake circuit is abnormal;	1. Check the brake circuit and power supply circuit;

Operation panel display	Small keyboard display	Fault description	Fault reason	Processing method
			2. The load is too large; 3. The encoder feedback signal is normal or not? 4. Is the motor parameter correct? 5. Check motor power line;	2. Reduce load; 3. Check whether the encoder feedback signal and the setting are correct, and whether synchronous motor encoder initial angle is correct; 4. Check motor parameters and do self-learning again;
Err 11	E11	Motor overload	1. Improper setting of FC-02; 2. The brake circuit is abnormal; 3. The load is too large;	1. Adjust parameters to keep FC-02 as default; 2. Refer to the fault content of ERR10;
Err 12	E12	Input side missing phase	1. Input power asymmetry; 2. The drive control panel is abnormal;	Check whether the input side three-phase power supply is balanced and whether the power supply voltage is normal;
Err 13	E13	Output side missing phase	1. Main circuit output wiring is loose; 2. Motor is damaged;	1. Check whether the contact point of KMY contactor is normal; 2. Exclude motor fault;
Err 15	E15	Abnormal output side	1. Main circuit output wiring is loose; 2. The output side grounding has short circuit;	
Err 16	E16	Encoder signal check is abnormal	Encoder signal check is abnormal	1. Check whether the encoder type is correct; 2. Check whether the encoder wiring is reliable; 3. Check whether the connection of the PG card is correct; 4. Whether the grounding of the control cabinet and the traction machine is good;
Err 19	E19	Motor tuning fault	1. the motor cannot operate properly; 2. overtime of motor self-learning process; 3. synchronous machine rotating encoder is abnormal;	1. Check and correctly input motor parameters; 2. Check whether the motor lead wire and the main contact point of the KMY contactor have missing phase; 3. Check the rotary encoder wiring and confirm the correct number of pulses per turn; 4. Whether or not release the maintenance operation button before the synchronous machine completes self-learning;
Err 20	E20	Speed feedback fault	1. Does the rotary encoder type match; 2. Rotary encoder wiring error; 3. Motor parameter setting error; 4. UVW power line phase error;	1. Check and confirm whether F1-00 is set correctly; 2. Check the encoder wiring is correct; 3. Check whether the brakes or other mechanical components are jammed during the operation of the elevator; 4. Change UVW phase power line of the motor;
Err 22	E22	Abnormal leveling signal	Sub-code 101: In floor switching process, the leveling signal is effective. Sub-code 102: From the elevator start up to	1. Please check the leveling and door area inductors are working normally; 2. Check the installation position of the leveling insert board is correct; 3. Check the input point of the motherboard level signal;

Operation panel display	Small keyboard display	Fault description	Fault reason	Processing method
			the floor switch, the falling edge of the leveling signal is not received. Sub-code 103: When the elevator is running automatically, the leveling position deviation is too large;	4. Check whether the slippage of wire rope is too big;
Err 24	E24	RTC clock signal fault	Abnormal controller clock signal;	Check and change the clock battery on the motherboard;
Err 26	E26	Seismic signal	Effective input of seismic signals;	Check seismic input signals and setting relevant parameters
Err 29	E29	Abnormal work of KMBF brake auxiliary contactor	When driving the synchronous machine, the feedback of KMBF brake auxiliary contactor is abnormal	1. Check whether the contactor feedback contact point is consistent with the main control panel parameter setting; 2. Check whether the indicator of the output part of the main control panel is consistent with the action of the contactor; 3. After checking the contactor action, whether the corresponding feedback contact acts, and whether the main control panel's corresponding feedback input point action is correct; 4. Check whether the output signal of KMBF contactor is consistent with that of main control panel;
Err 30	E30	Elevator position anomaly	1. When the elevator operates automatically, the position of the feedback of the rotary encoder has deviation; 2. During elevator's automatic operation, the leveling signal disconnects or there is adhesion phenomenon; 3. Wire rope slipping or motor blocking;	1. Check whether there is false action of the leveling inductor in the non leveling area; 2. Check whether the leveling signal connection is reliable, and whether it is possible for grounding, or short circuit with other signals; 3. Check whether the encoder installation is in place;
Err 33	E33	Elevator speed anomaly	Sub-code 101: Over speed of rated speed operation; Sub-code 102: Over speed of inspection or shaft self-learning operation; Sub-code103: Over speed of sliding operation; Sub-code 104: Over speed of emergency operation;	1. Confirm whether the rotary encoder type is correct or not; 2. Check that motor parameter setting is correct; 3. Carry out motor self-learning again;
Err 35	E35	Shaft self-learning data anomaly	Sub-code 101: When self-learning starts, the current floor is not the minimum floor or the next level forced deceleration is invalid.	101 fault: Check whether the next level forced deceleration is effective; whether the current floor F4-01 is the lowest floor;

Operation panel display	Small keyboard display	Fault description	Fault reason	Processing method
			Sub-code 102: The inspection switch disconnects during the shaft self-learning. Sub-code 103: The system is powered on and finds that the shaft self-learning is not conducted. Sub-code 104: In distance control mode, the self-learning of the shaft is not conducted when the operation is started. Sub-code 105: The direction of the elevator operation and that of pulse change is inconsistent Sub-code 106, 107, 109, 114: Abnormal leveling board pulse length induced by up and down leveling; Sub-code 108 and 110: Self-learning leveling signal does not change over 45S. Sub-code 111 and 115: The stored floor height is less than 50CM Sub-code 112: Before the self-learning is completed, the current layer is not the top layer Sub-code 113: Abnormal pulse check	102 fault: Check whether the elevator is in inspection mode 103, 104 fault: Carry out shaft self-learning again 105 fault: Please confirm whether the change of the elevator operation is the same as the pulse change of F4-03, and if they are not consistent, please adjust by F2-10. 106, 107, 109, and 114 fault: The normal open and normal close setting of the leveling inductor is incorrect; The leveling inductor signal is flashing, and please check whether the levelling board is installed in place, and whether there is strong current interference; 108, 110 fault: The operating time exceeds the time for protecting F9-02, and the system still has not received the leveling signal; 111, 115 faults: If the floor height is less than 50CM, please open the ultra short floor function; if not, please check the leveling board installation of this floor, or check the inductor; 112 fault: The setting value of the maximum floor F6-00 does not match the actual performance; 113 fault: Check whether the leveling inductor signal is normal, and conduct the shft self-learning again;
Err 36	E36	Abnormal feedback of KMY operation contactor	1. When the elevator starts, the running contactor has output, but there is no feedback signal; 2. During the elevator operation, the operation contactor feedback signal is lost; 3. Adhesion of contactor feedback signal;	1. Check whether the feedback contact action of the contactor is normal; 2. Check whether the main board parameter is set correctly;
Err 37	E37	Abnormal feedback of	The brake output is not consistent with the feedback	1. Check whether the brake contactor feedback contact

Operation panel display	Small keyboard display	Fault description	Fault reason	Processing method
		KMB brake contactor	signal;	action is normal; 2. Check whether the main board parameter is set correctly;
Err 38	E38	Signal abnormality of controller rotary encoder	1. When the elevator is running automatically, the direction of the rotary encoder signal input is not correct; 2. Set to open loop operation (F0-00) under distance control;	1. Determine the parameter setting value of F0-00 as "1"; 2. Change the parameters of the elevator running direction F2-10;
Err 39	E39	Overheating fault of motor	The motor's overheating signal input is effective;	1. Check whether the motor is damaged; 2. Check whether the motor thermal switch action is normal;
Err 41	E41	Abnormal safety circuit disconnection	Disconnection of safety circuit signal;	1. Check safety circuit's safety switch; 2. Check whether external power supply is correct;
Err 42	E42	Abnormal disconnection of door lock circuit	During the elevator operation, the feedback signal of the door lock circuit is disconnected;	1. Check whether the door locks of landing door and car door are normally contacted; 2. Check the door lock feedback signal is normal; 3. Check that the peripheral power supply is normal;
Err 43	E43	Abnormal up limit signal during operation	The up limit signal is disconnected during the up running of the elevator;	1. Check whether the up limit signal is normal; 2. Check whether the up limit switch is in normal contact; 3. Check the installation position of the up limit switch;
Err 44	E44	Abnormal down limit signals during operation	The down limit signal is disconnected during the down running of the elevator;	1. Check whether the down limit signal is normal; 2. Check whether the down limit switch is in normal contact; 3. Check the installation position of the down limit switch;
Err 45	E45	Forced deceleration switch abnormality	1. Forced deceleration signal abnormality; 2. During the shaft self-learning process, the installation position of the forced deceleration switch is not correct;	1. Check whether the up and down level 1 deceleration switch is in normal contact; 2. Check the installation distance of the forced deceleration switch is correct;
Err46	E46	Re-leveling abnormality	Sub-code 101: The leveling signal is invalid in the re-leveling operation; Sub-code 102: The re-leveling speed is more than 0.1m/s; Sub-code 103: When the normal operation starts, the detection signal of the re-leveling is abnormal; Sub-code 104: The re-leveling operation did not receive the door closing feedback signal 2S after the output of door closing;	101 fault: Check whether the leveling signal is normal; 102 fault: Confirm whether the use of rotary encoder is correct; 103, 104 fault: a. Check whether the leveling sensor signal is normal; b. Check the door closing feedback input point; c. Check the SMZ security module and wiring;
Err 47	E47	Abnormal adhesion of door closing	During the pre-opening and re-leveling, the door closing contactor appears adhesion	1. Check input signal of door closing contactor feedback; 2. check whether the door closing

Operation panel display	Small keyboard display	Fault description	Fault reason	Processing method
		contactor	phenomenon;	contactor action is normal;
Err 48	E48	Door opening fault	The number of continuous door opening exceeds the set value of FB-13, but the door lock is not opened;	1. Check whether the door machine system works normal; 2. Check car top control panel works properly;
Err 49	E49	Door closing fault	The number of continuous door closing exceeds the set value of FB-13, but the door lock is not closed;	1. Check whether the door machine system is normal; 2. Check car top control panel works properly;
Err 50	E50	Continuous loss of leveling signal	Adhesion and loss of leveling signal for 3 times continuously (that is, report E22 for three times continuously)	1. Check whether the leveling inductor is working normally; 2. Check the installation depth of the leveling board; 3. Checking whether there is slippage procedure for wire rope;
Err 51	E51	Abnormal CAN communication	CAN serial communication error;	1. Check whether the connection of the communication cable is correct; 2. check the roof control panel power supply and work is normal;
Err 52	E52	Abnormal hall calling communication	Hall calling serial communication error;	1. Check the communication cable connection is correct; 2. Check the address of the hall calling control board is set correctly;
Err 53	E53	Door lock short circuit fault	When the elevator is in automatic operation, the door lock is not disconnected after the stop;	1. Check whether the door lock action is normal; 2. Check whether the door lock feedback signal is normal; 3. Check whether the door opening and closing in place signal is normal;
Err 54	E54	Over current during inspection operation start-up	When in inspection mode, the starting current is higher than 110% of the rated current	1. Load is too large, and reduce load; 2. Temporarily change FC-00 to 2 (10);
Err 55	E55	Fault of floor change and stop	Under automatic operation mode, the door not open at the current floor;	Check whether the door lock appears short circuit;
Err 57	E57	Abnormal SPI communication	SPI communication fault;	Check whether the connection between the main control board and the drive panel is reliable;
Err 58	E58	Abnormality of shaft position protection switch	1. The up / down forced deceleration switch is abnormal; 2. The up / down limit switch is abnormal;	1. Check the forced deceleration switch and related parameters; 2. Check the limit switch and related parameters;
Err 65	E65	The unintended car movement is detected	1. unintended movement of car; 2. Working brake isn't closed fully; 3. The leveling inductor and installation distance is abnormal; 4. SMZ module signal is abnormal;	1. Check working brake; 2. Check leveling inductor; 3. Check SMZ security module;
Err 66	E66	The fault detection of the braking force	1. Insufficient braking force of working brake; 2. Wearing of brake shoe or foreign body on the surface;	1. Check working brake; 2. Check the parameter values set in the controller;

Operation panel display	Small keyboard display	Fault description	Fault reason	Processing method
			The detection torque inside the system is too large;	
Err 67	E67	Door lock is not opened	Elevator arrives at the stop, but door lock is not opened	1. Check whether there are foreign objects on the sill; 2. Check mechanical lock hook distance of the door lock;
Err 68	E68	The door opening in place signal cannot be received	Elevator has not received the door opening in place signal at 3 different floors during operation	1. Check whether there are foreign objects on the sill; 2. Check mechanical lock hook distance of the door lock; 3. Check door opening in place signal of door machine;
Err 69	E69	Stuck phenomenon of hall call button	Abnormal hall call button	Check the hall call button
Err 70	E70	Stuck phenomenon of button in the car	Abnormal button in the car	Check call button in car
Err 80	E80	Abnormal feedback signals of auxiliary brake	Abnormal action of auxiliary brake	1. Check the movement flexibility of additional brake; 2. Check action detection switch of the auxiliary brake;
Err 88	E88	E88 is the prompt information used for the simulation test of car unintended movement protection		



The fault code, the floor signal and fault sub-code will be scrolling displayed on the mainboard digital tube when the elevator has any fault. For example, "E42-L02-101" means that the elevator fault code is E42, the elevator fault is happened on the second floor, and the fault sub-code is 101!

Appendix 4 Adjustment of Door Machine Parameters

1. AAD03010 Series Door Machine Control System:

Parameters of P area: P01 ~ P79:

Function code	Function description	Unit	Setting scope	Factory setting value	Note
P01	The first acceleration time	s	0~999	0.5	▲
P02	The first deceleration time	s	0~999	0.5	▲
P03	V/F method (maximum output frequency)	Hz	50、60	50	※
P04	V/F curve		0~1	0	※
P05	Torque increase	%	0~40	18	▲
P06	Select electronic thermal function mode		0~3	2	※
P07	Set thermistor current	A	0.1~100	2.8	※
P08	Operation instruction selection		0~3	2	※
P09	Frequency setting signal		0~6	4	※
P10	Stop mode		0~1	0	※
P11	Stop frequency	Hz	0.5~60	0.5	※
P12	DC braking time	s	0~120	0	※
P13	DC braking level		0~100	0	※
P14	Maximum output frequency	Hz	50~250	50	※
P15	Base frequency	Hz	45~250	50	※
P16	Prevent over current stall function		0~1	1	※
P17	Prevent over voltage stall function		0~1	1	※
P22	Current limit function	s	0~9.9	0	※
P23	Start-up mode		0~3	2	※
P24	Choose sudden stop and re-start		0~2	0	※
P25	Standby time	s	0.1~100	0.1	※
P26	Choose re-trial		0~3	0	※
P27	Re-trial times		1~10	1	※
P28	Lower limit frequency	Hz	0.5~250	0.5	※
P29	Upper limit frequency	Hz	0.5~250	250	※
P30	Monitoring selection		0~1	0	※
P32	Maximum output voltage	V	0~500	0	※
P33	OCS electrical level	%	1~200	70	※
P34	Carrier frequency	kHz	0.8~15.0	12.5	▲
P42	Clear setting data		0~2	0	▲ when setting to 1 it is factory reset value
P43	set input signal logic		0~31	0	※
P44	Response time of safety sensor	ms	0~999	10	※
P45	Response time of arrival signal	ms	1~999	10	※
P47	Detection time of arrival SW abnormality	s	0~10	0	※
P48	RY1 function selection		0~7	7	▲
P49	RY2 function selection		0~7	4	※
P50	RY3 function selection		0~7	5	※
P59	Overload detection frequency 1	Hz	0.5~250	10.0	※
P60	Overload detection frequency 2	Hz	0.5~250	50.0	※
P61	Overload detection current 1	A	0.1~100	1.2	※
P62	Overload detection current 2	A	0.1~100	1.2	※

Function code	Function description	Unit	Setting scope	Factory setting value	Note
P63	Overload detection determination time	ms	0、1~999	0	※
P64	Clipping in determination frequency ratio (low speed)	%	0~100	50.0	
P65	Clipping in determination frequency ratio (high speed)	%	0~100	70.0	
P66	Clipping in determination switching frequency	Hz	0.5~250	5.0	
P67	Clipping in detection determination time	ms	0~999	100	
P68	Start-up determination time	ms	100~999	200	※
P69	Determination time of forced door opening action	s	0~500	0	※
P70	Determination time of abnormal door opening action	s	0~500	0	※
P71	Holding time of abnormal door opening action arrival	s	0.0~10	1	※
P72	Holding time of repeated door opening arrival	s	0.0~10	3.0	
P73	Holding time of repeated door closing arrival	s	0.0~10	3.0	
P74	S acceleration and deceleration function		0、1、2	1	※
P75	Stop mode selection in door opening / closing action		0、1	0	※
P76	No arrival signal selection		0、1、2	0	



“※” indicates that modifying this parameter may cause errors in the system, and it is recommended not to modify it at jobsite;

“▲” indicates that the parameter can be adjusted according to actual situation at jobsite;

Parameters in d area: d01-d53:

Function code	Function description	Unit	Setting scope	Factory setting value	Note
d14	Door closing arrival holding frequency (door closing holding force)	Hz	0、0.5~250	0.5	▲
d15	Door opening frequency 1 (door opening start-up speed)	Hz	0、0.5~250	20/6	▲ (set to 6 in three loop motor)
d17	Door opening frequency 3 (door opening high speed)	Hz	0、0.5~250	25/19	▲ (set to 19 in three loop motor)
d20	Door opening frequency 6 (door opening low speed)	Hz	0、0.5~250	8/5	▲ (set to 5 in three loop motor)
d21	Door opening arrival holding frequency (door opening holding force)	Hz	0、0.5~250	2	▲
d22	Door closing frequency 1 (door closing start-up speed)	Hz	0、0.5~250	20	▲
d24	Door closing frequency 3 (door closing high speed)	Hz	0、0.5~250	30	▲
d27	Door closing frequency 6 (door closing low speed)	Hz	0、0.5~250	3/1.8	▲ (set to 1.8 in three loop motor)
d28	Door opening acceleration and deceleration time 1 (Door opening start-up acceleration time)	s	0、0.1~999	0.5/2	▲ (set to 2 in three loop motor)

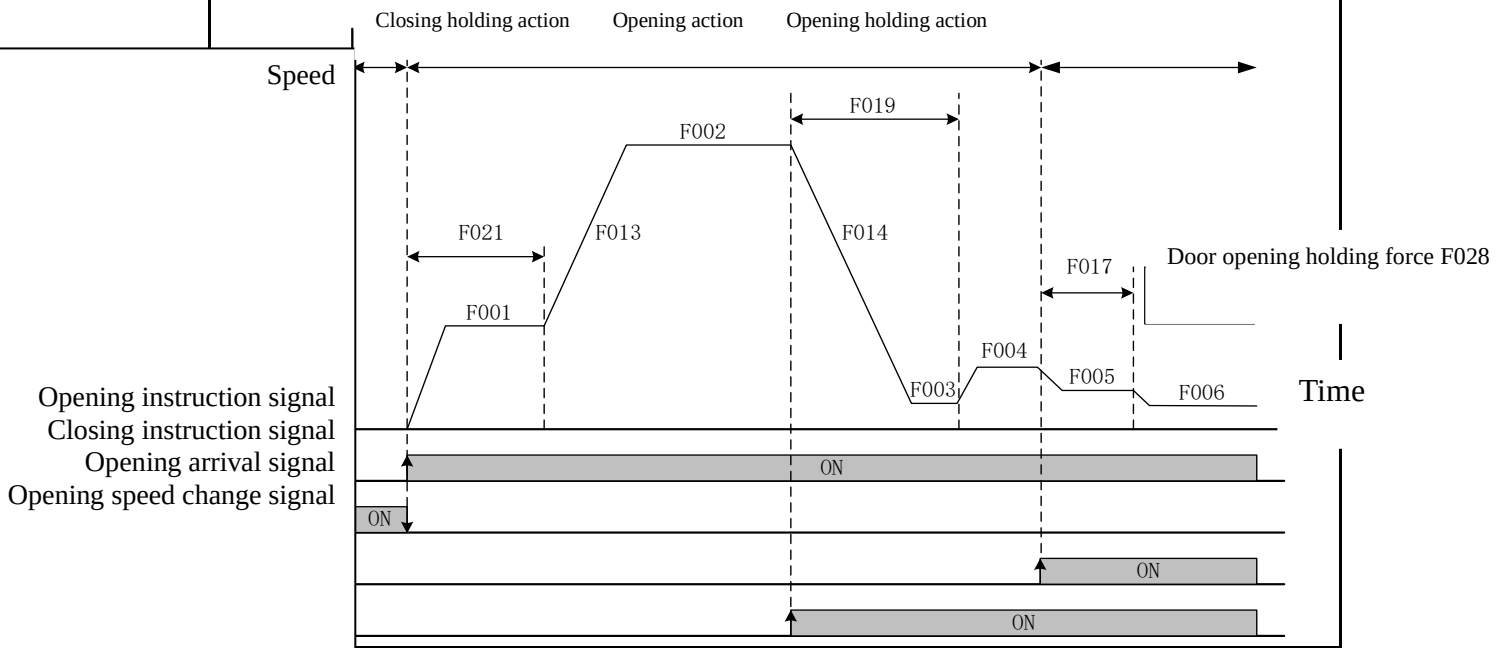
Function code	Function description	Unit	Setting scope	Factory setting value	Note
d30	Door opening acceleration and deceleration time 3 (Door opening high speed acceleration time)	s	0、0.1~999	2.2	▲
d33	Door opening acceleration and deceleration time 6 (Door opening deceleration time)	s	0、0.1~999	0.8/1.3	▲ (set to 1.3 in three loop motor)
d34	Door closing acceleration and deceleration time 1 (Door closing start-up acceleration time)	s	0、0.1~999	1	▲
d36	Door closing acceleration and deceleration time 3 (Door closing high speed acceleration time)	s	0、0.1~999	0.5	▲
d39	Door closing acceleration and deceleration time 6 (Door closing deceleration time)	s	0、0.1~999	1.2/2.3	▲ (set to 2.3 in three loop motor)
d40	Holding current in door opening	A	0、0~100	0.4/0.2	▲ (set to 0.2 in three loop motor)
d41	Holding current in door closing	A	0、0~100	0.4/0.2	▲ (set to 0.2 in three loop motor)
d42	Stop time of door opening/ closing holding action	s	0~999	0	※
d43	Holding standby frequency in door opening arrival	Hz	0.5~250	2	▲
d44	Holding standby frequency in door closing arrival	Hz	0.5~250	2	▲
d45	Holding standby time in door opening arrival	s	0.0~10	0.5	▲
d46	Holding standby time in door closing arrival	s	0.0~10	0.5	▲
d47	Door opening initiation timer (SW method)	s	0.0~3.0	0	※
d48	Door opening deceleration timer (SW method)	s	0.0~3.0	0	※
d49	Door closing initiation timer (SW method)	s	0.0~3.0	0	※
d50	Door closing deceleration timer (SW method)	s	0.0~3.0	0	※
d51	Action frequency under the power supply ON condition	Hz	0、0.5~250	0	※



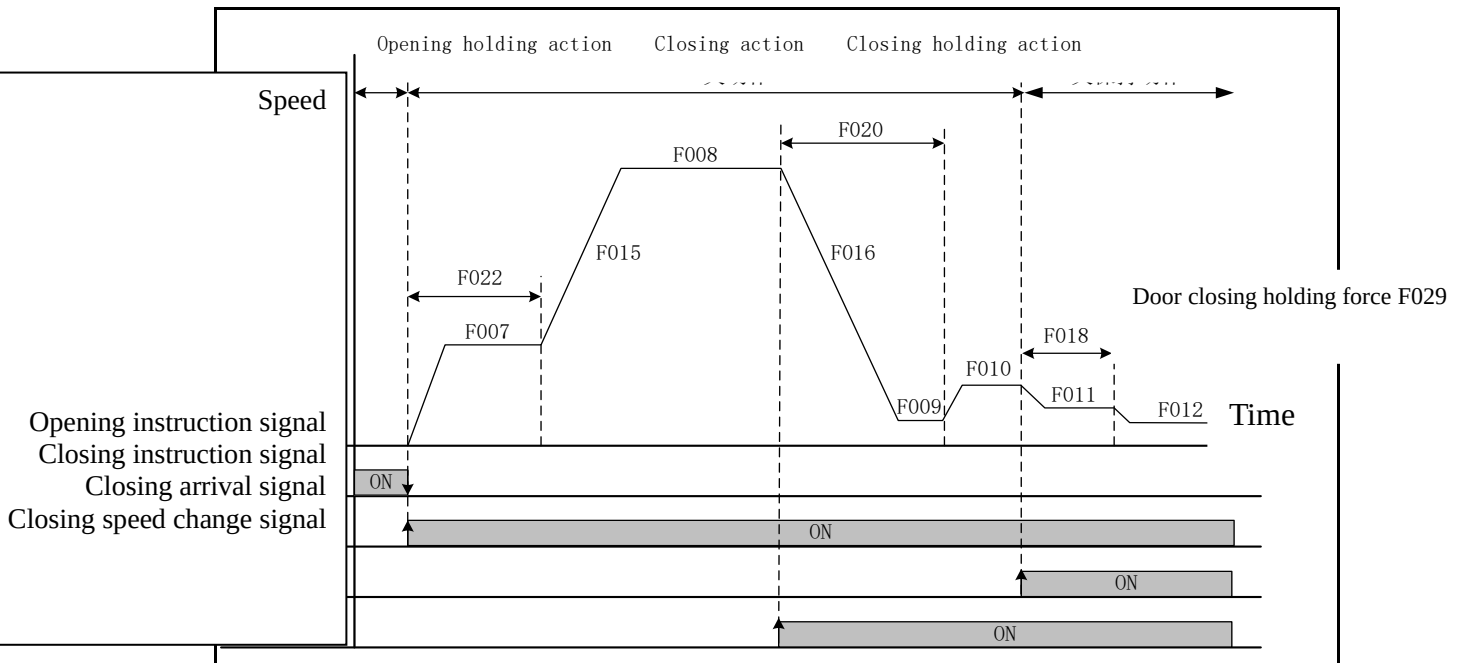
“※” indicates that modifying this parameter may cause errors in the system, and it is recommended not to modify it at jobsite;

“▲” indicates that the parameter can be adjusted according to actual situation at jobsite;

2. LB20GMD Series Door Machine Control System:



(Door opening sequence diagram)



(Door closing sequence diagram)

Control method and motor parameters:

Code	Function	Description	Setting scope	Value at factory
F000	Door machine operation mode	0-automatic repeated door opening and closing mode 1-elevator switch mode	0~1	※ 1

F001	Door opening starting speed	Multi-section speed of door opening process, see the door opening process sequence diagram in detail	0~50Hz	▲ 10
F002	Door opening high speed			▲ 22
F003	The first low speed of door opening			▲ 7
F004	The second low speed of door opening			▲ 7
F005	Door opening in place low speed			▲ 4
F006	Door opening holding speed			1
F007	Door closing starting speed	Multi-section speed of door closing process, see the door closing process sequence diagram in detail	0~50Hz	▲ 10
F008	Door closing high speed			▲ 20
F009	The first low speed of door closing			▲ 4
F010	The second low speed of door closing			▲ 4
F011	Door closing in place low speed			▲ 3
F012	Door closing holding speed			0.5
F013	Door opening acceleration time	The starting time from 0Hz to 50Hz	0.1~ 100 s	▲ 1.5
F014	Door opening deceleration time	The stopping time from 50Hz to 0Hz		▲ 1.2
F015	Door closing acceleration time	The starting time from 0Hz to 50Hz		▲ 1.5
F016	Door closing deceleration time	The stopping time from 50Hz to 0Hz		▲ 1.2
F017	Door opening in place low speed operating time			▲ 2.0
F018	Door closing in place low speed operating time			▲ 2.0
F019	Operating time of the first low speed of door opening	F003 operatin time (including F014)		▲ 1.2
F020	Operating time of the first low speed of door closing	F004 operatin time (including F016)		▲ 1.2
F021	Door opening starting speed operating time			0.5
F022	Door closing starting speed operating time			0.5
F026	Intermittent stop time after automatic repeated door opening and closing mode in place		0.1~ 100 s	5.0
F027	Stop time of door opening/closing holding action		0.1~ 100 s	0.0
F028	Door opening holding force		0~40%	▲ 15
F029	Door closing holding force			▲ 15
F030	Door re-opening deceleration time		0.1~ 100 s	0.5
F031	Door re-opening acceleration time			0.5
F032	Starting torque increasement	The torque compensation of door opening and closing starting process	0~40%	▲ 15

F033	Door opening and closing action time limit	Automatically carry out re-opening when exceeding this time during the door closing process	0.1~ 100 s	8.0
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※ -indicates that the factory can modify the function and performance setting of the product but the users shall not modify them;

▲ -indicates that the user can modify according to the actual operation effect of the door machine at jobsite.

3. LB600GM Series Door Machine Control System:

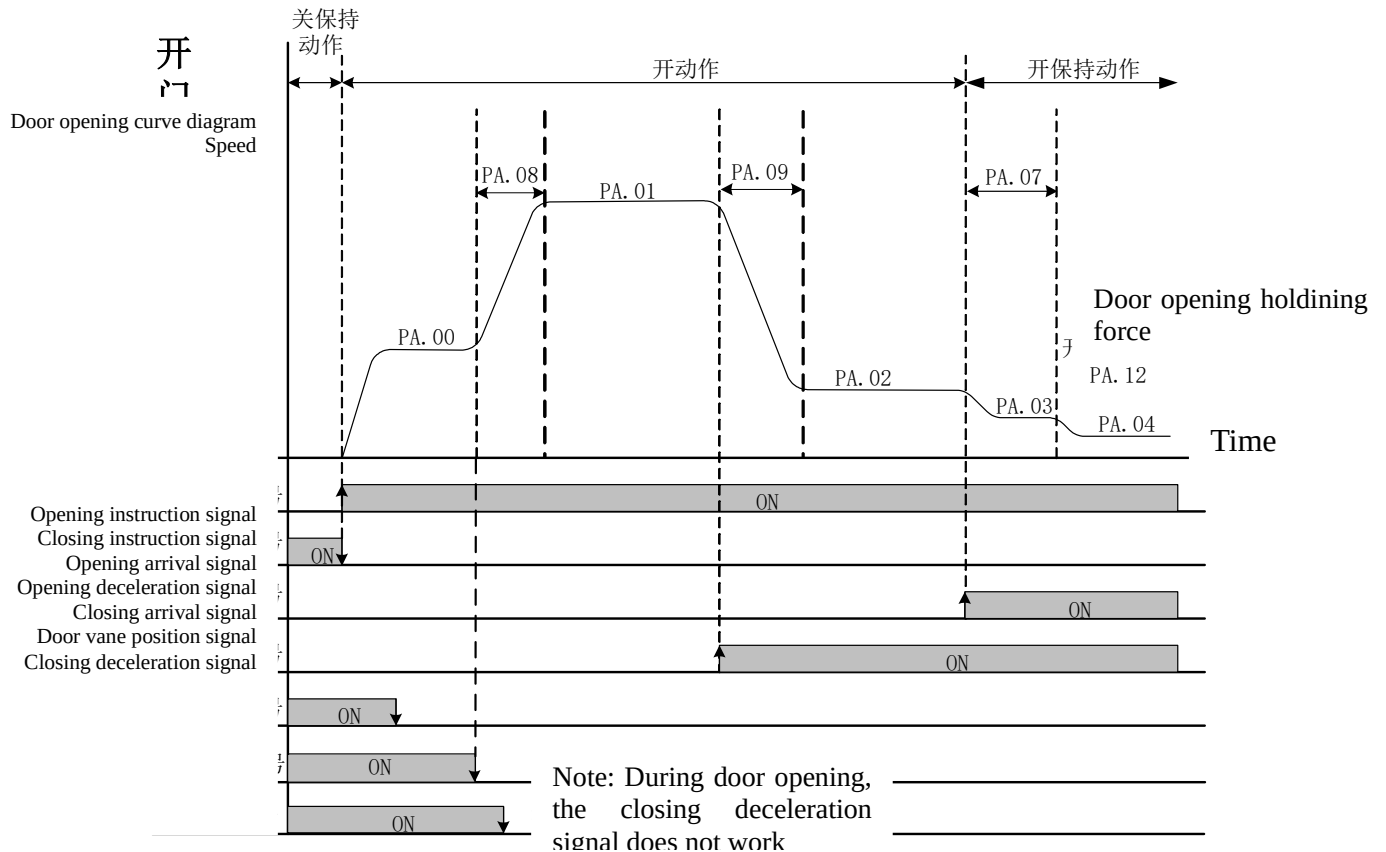
3.1 Door machine frequency converter parameters:

Functio	Name	Setting scope
P0.00	Control mode	2: Closed loop vector control
P2.00	Motor type	1: Synchronous motor
P2.03	Motor's rated current	Set according to the motor
P2.04	Motor's rated frequency	Set according to the motor
P2.05	Motor's rated speed	Set according to the motor

3.2 Speed curve:

(1) Door opening speed curve

Closing holding action Opening action Opening holding action

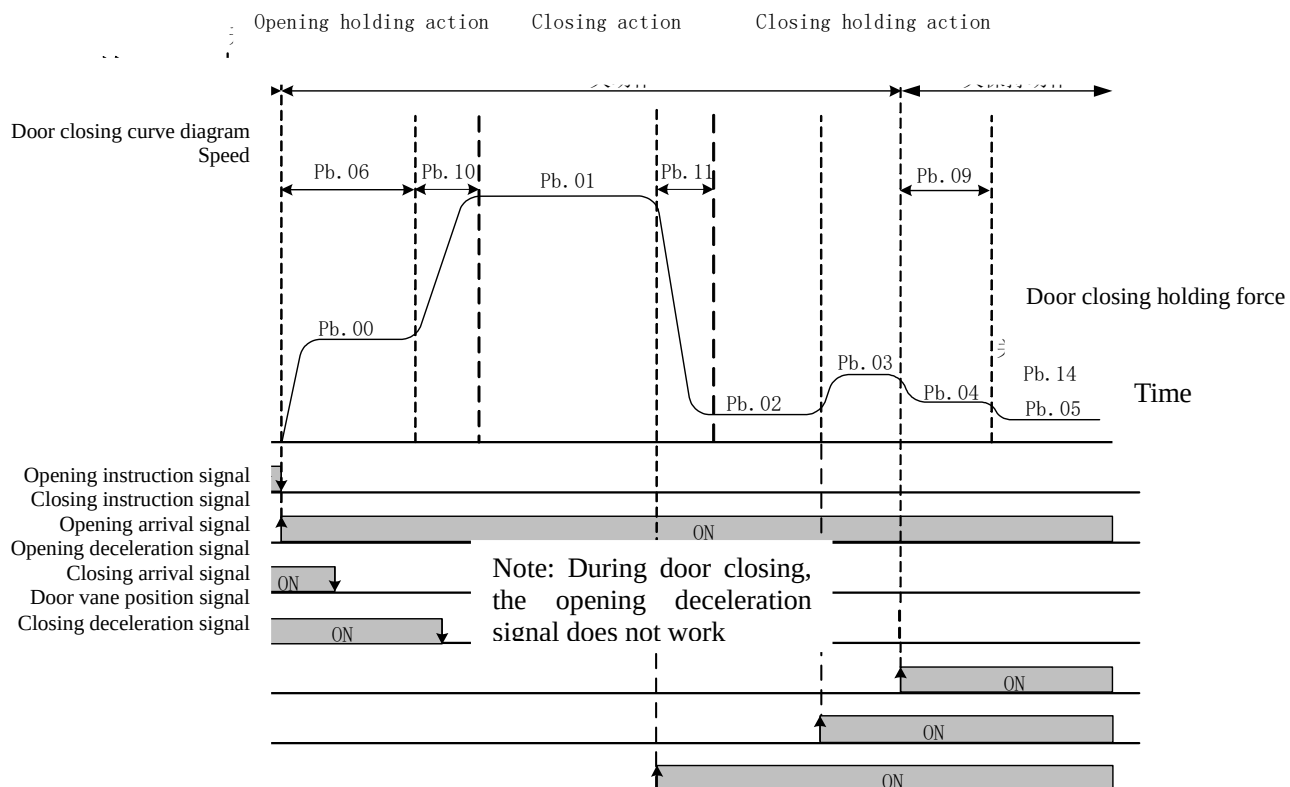


Note: The acceleration and deceleration time of the whole door opening process shall be controlled by PA.08 and PA.09!

(2) Door opening parameter group

Funcio	Name	Setting scope
PA.00	Door opening starting speed	0.00 Hz ~ P0.04
PA.01	Door opening high speed	0.00 Hz ~ P0.04
PA.02	Door opening low speed	0.00 Hz ~ P0.04
PA.03	Door opening in place low speed	0.00 Hz ~ P0.04
PA.04	Door opening holding speed	0.00 Hz ~ P0.04
PA.07	Door opening in place low speed operating time	0.0~100.0 S 【1.0】
PA.08	Door opening acceleration time	0.1~100.0 S 【2.8】
PA.09	Door opening deceleration time	0.1~100.0 S 【2.9】
PA.10	Door opening torque	5.0~150.0% 【150.0】
PA.11	Door opening holding torque	5.0~150.0% 【50.0】
PA.13	Door opening in place switching torque	5.0~150.0% 【60.0】
PA.14	Door re-opening deceleration time	0.1~100.0 S 【1.0】
PA.15	Door re-opening acceleration time	0.1~100.0 S 【2.5】

(3) Door closing speed curve



Note: The acceleration and deceleration time of the whole door closing process shall be controlled by Pb.10 and Pb.11!

(4) Door closing parameter group

Function	Name	Setting scope
Pb.00	Door closing starting speed	0.00 Hz ~ P0.04 (maximum frequency) [4]
Pb.01	Door closing high speed	0.00 Hz ~ P0.04 (maximum frequency)
Pb.02	The first low speed of door closing	0.00 Hz ~ P0.04 (maximum frequency)
Pb.03	The second low speed of door closing	0.00 Hz ~ P0.04 (maximum frequency)
Pb.04	Door closing in place low speed	0.00 Hz ~ P0.04 (maximum frequency)
Pb.05	Door closing holding speed	0.00 Hz ~ P0.04 (maximum frequency)
Pb.06	Door closing starting speed operating time	0.0~100.0 S [0.5]
Pb.09	Door closing in place low speed operating time	0.0~100.0 S [1.0]
Pb.10	Door closing acceleration time	0.1~100.0 S [2.8]
Pb.11	Door closing deceleration time	0.1~100.0 S [3.2]
Pb.12	Door closing torque	5.0~150.0% [150.0]
Pb.13	Door closing holding torque	5.0~150.0% [50.0]
Pb.15	Door closing blocked action	0~2 [1]

Function	Name	Setting scope
Pb.16	Opening and closing door blocking determination time	0.1~2.0 S [0.3]
Pb.17	Closing door in place switching torque	5.0~150.0% [60.0]
Pb.18	Forced door closing high speed	0.00 Hz~P0.04(maximum frequency)

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